

Artificial Intelligence-Driven Cognitive Telerehabilitation for Stroke: A Randomized Controlled Trial

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Background and Aims

Cognitive impairment is a common consequence of stroke, requiring effective rehabilitation strategies. Telerehabilitation has emerged as a promising alternative to in-person cognitive therapy, yet existing systems often lack mechanisms for real-time personalization and engagement monitoring. This study aimed to evaluate the clinical effectiveness of a self-guided AI-driven cognitive telerehabilitation (Zenicog®, Figure 1) compared to a therapist-supervised rehabilitation in subacute stroke patients.

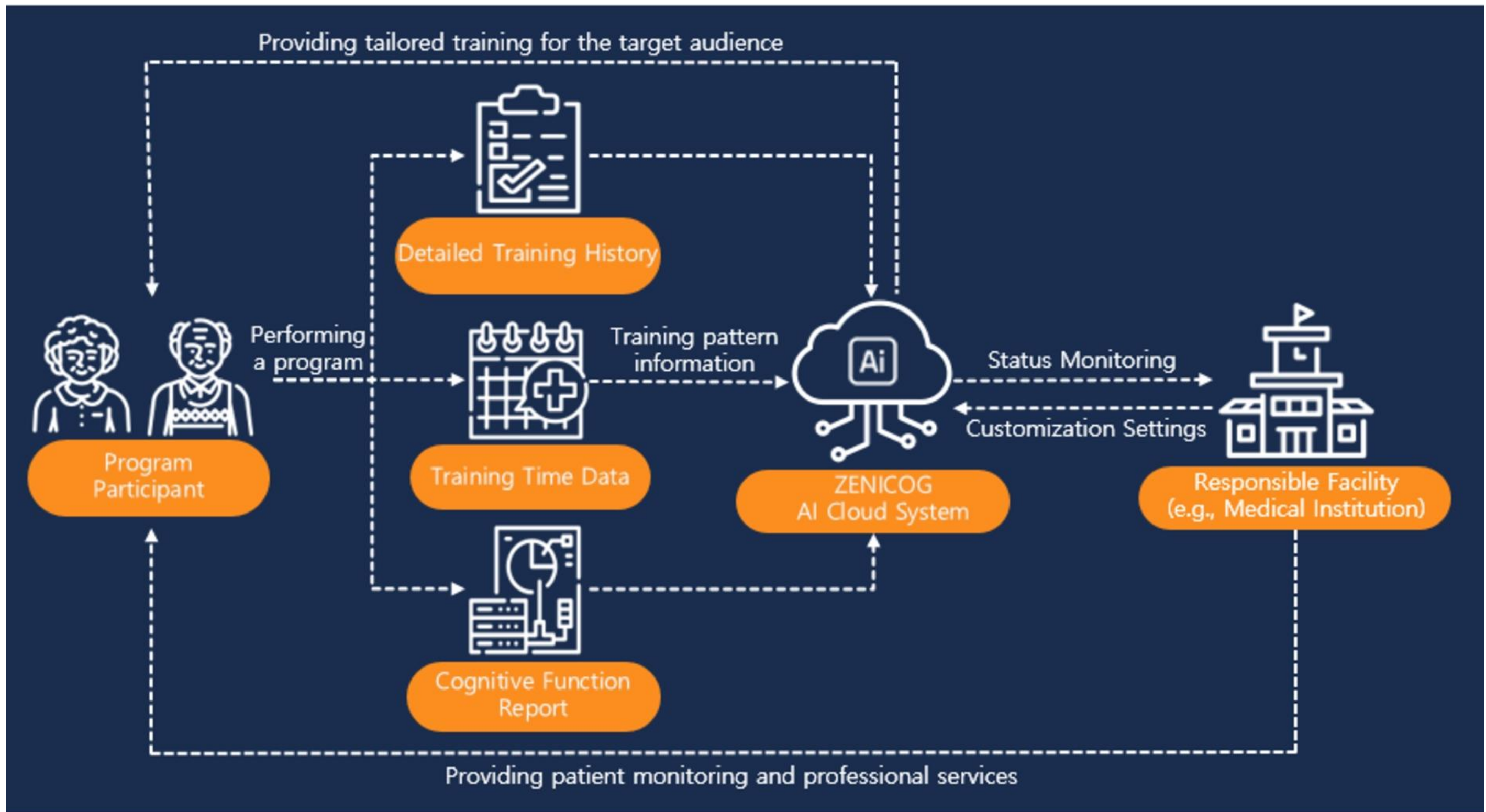


Figure 1. Schematic overview of the Zenicog® for AI-based cognitive rehabilitation.

Methods

In this multicenter, parallel-group, randomized controlled non-inferiority trial, 63 participants with cognitive impairment within six months of stroke onset were randomized 1:1 to either a self-guided AI-driven telerehabilitation group or a therapist-supervised rehabilitation group. Both groups completed 24 sessions within six weeks using the same mobile platform. The primary outcomes were cognitive function measures, including the Korean Mini-Mental State Examination-2 (K-MMSE2), Trail Making Tests (A and B), and Digit Span Tests (forward and backward), with non-inferiority formally tested using the K-MMSE2. Secondary outcomes included functional independence, psychosocial measures and usability questionnaire.

Table 1. Within-Group Comparison of Pre- and Post-Intervention Changes

	Total			Intervention			Control		
	pre	post	P-value	pre	post	p-value	pre	post	P-value
MMSE	22 [15;25]	24 [19;27]	<0.001*	23 [17;25]	26 [21;27]	<0.001*	19 [14;23]	23 [18;26]	<0.001*
DSF	5 [4;6]	5 [4;7]	0.016*	6 [4;7]	6 [4;7]	0.007*	4 [4;6]	5 [4;6]	0.226
DSB	2 [2;3]	3 [2;4]	0.009*	3 [2;3]	3 [2;4]	0.015*	2 [2;3]	2 [2;3]	0.183
TMT_A	61.58 [33.69;117.94]	53.59 [31.94;74.46]	0.001*	48.58 [30.13;91.65]	47.39 [23.8;72.43]	0.011*	68.54 [45.40;164.49]	64.24 [39.42;75.82]	0.026*
TMT_B	134.63 [73.87;300]	102.22 [67.8;168.31]	<0.001*	113.94 [70.65;257.46]	81.25 [47.83;141.19]	0.001*	152.44 [80.89;300]	109.69 [79.56;266.33]	0.014*
CES-D	16 [8;32]	14 [5;25]	0.037*	14 [7;24]	10 [4;20]	0.111	23.0 [9.0;34.5]	18 [7;33]	0.161
EQ-5D-5L	13.89 ± 5.11	13.33 ± 4.94	0.299	13.11 ± 5.18	13 ± 5.17	0.877	14.64 ± 5.01	13.64 ± 4.79	0.226
SES	28 [25;31]	30 [24;32]	0.477	30 [27;33]	30 [26;36]	1	26 [22;29]	29 [22;31]	0.353
MBI	47.67 ± 22.40	55.64 ± 20.79	<0.001*	49.59 ± 23.12	57.19 ± 21.33	<0.001*	45.8 ± 21.9	54.1 ± 20.5	<0.001*

Values, mean ± SD or median [1st quartile, 3rd quartile]; MMSE, Mini-Mental State Examination; DSF, Digit Span Forward; DSB, Digit Span Backward; TMT_A, Trail Making Test Part A; TMT_B, Trail Making Test Part B; CES-D, Center for Epidemiologic Studies Depression Scale; EQ-5D-5L, EuroQol 5-Dimension 5-Level Questionnaire; SES, Socioeconomic Status; MBI, Modified Barthel Index

Results

Fifty-five participants completed the study. Both groups showed significant improvements across all primary cognitive measures (Table 1), with no statistically significant differences between groups (Table 2). Non-inferiority analysis confirmed that the self-guided AI-driven telerehabilitation was not inferior to the therapist-supervised rehabilitation based on K-MMSE2 changes (Figure 2). Usability assessment among users of the cognitive rehabilitation system indicated high overall satisfaction with no serious adverse events reported.

Table 2. Comparison of Treatment Effectiveness Between the Two Groups

	Total (N = 55)	Intervention (N = 27)	Control (N = 28)	p-value
MMSE	2.76 ± 2.91	2.48 ± 2.81	3.04 ± 3.04	0.486
DSF	0 [0;1]	0 [0;1]	0 [0;1]	0.614
DSB	0 [0;1]	0 [0;1]	0 [0;1]	0.536
TMT_A	-5.29 [-19.27;0]	-5 [-15.16;0]	-5.73 [-21.09;0]	0.846
TMT_B	-10.91 [-39.16;0]	-17.09 [-39.16;-2.89]	-5.43 [-36.87;0]	0.272
CESD	-3.56 ± 12.37	-3.59 ± 13.83	-3.54 ± 11.04	0.987
SES	0 [-3;3]	0 [-3;2]	1 [-3;4]	0.510
MBI	7 [0;15]	3 [0;13]	9 [0;13]	0.416

Values, mean ± SD or median [1st quartile, 3rd quartile]; MMSE, Mini-Mental State Examination; DSF, Digit Span Forward; DSB, Digit Span Backward; TMT_A, Trail Making Test Part A; TMT_B, Trail Making Test Part B; CESD, Center for Epidemiologic Studies Depression Scale; EQ-5D-5L, EuroQol 5-Dimension 5-Level Questionnaire; SES, Socioeconomic Status; MBI, Modified Barthel Index

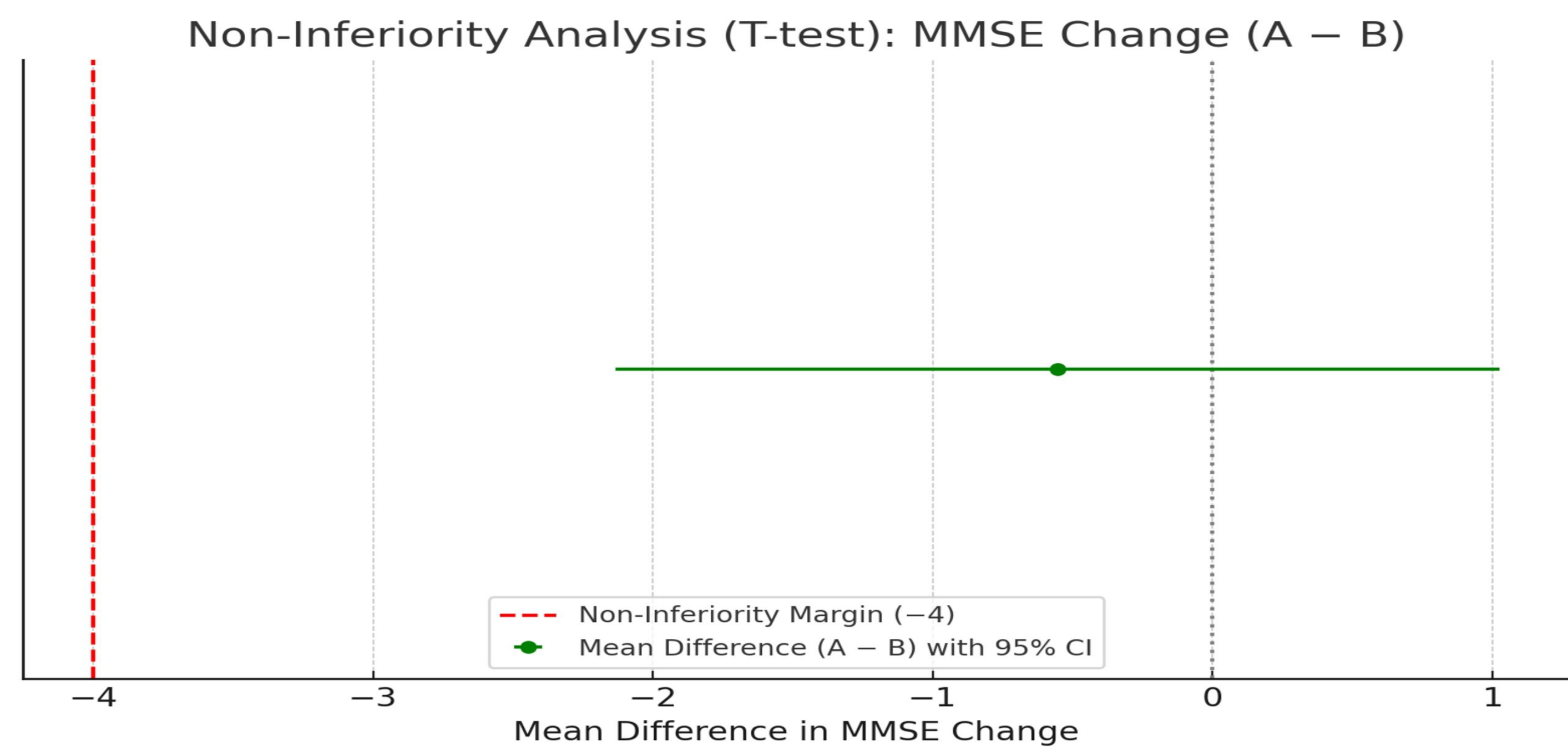


Figure 2. Non-inferiority analysis of the AI-based intervention compared to conventional therapy.

Conclusion

This study demonstrates that a self-guided, AI-driven telerehabilitation can deliver cognitive improvements comparable to a therapist-supervised rehabilitation in subacute stroke patients.

